

September 24, 2026

When running the scripts to reproduce the numerical experiments in the article, be aware that in the published version you can find the following typos (corrected in blue). Sorry for any inconvenience this may cause you as a reader. In this brief note, you can find the results obtained by running the code today.

r	$\max_{q \leq n \leq M} \ \mathbf{u}_r^n - \mathbf{u}_h(t_n)\ $ for different q				
	$q = 1$	$q = 2$	$q = 3$	$q = 4$	$q = 5$
18	1.2915×10^{-1}	1.8004×10^{-1}	1.7983×10^{-1}	1.7983×10^{-1}	1.7983×10^{-1}
26	5.3288×10^{-2}	1.5079×10^{-3}	1.3063×10^{-3}	1.3020×10^{-3}	1.3020×10^{-3}
34	4.2505×10^{-2}	7.9057×10^{-4}	6.5009×10^{-5}	5.9256×10^{-5}	5.9739×10^{-5}
42	3.8688×10^{-2}	7.7619×10^{-4}	2.6741×10^{-5}	3.1813×10^{-6}	2.2734×10^{-6}
50	6.3663×10^{-2}	1.0528×10^{-3}	2.9448×10^{-5}	2.4092×10^{-6}	3.0961×10^{-7}

Table 3: Maximum errors in L^2 norm for $1 \leq q \leq 5$ and different values of r .

	$q = 2$	$q = 3$	$q = 4$	$q = 5$
$M = 64$	5.647×10^{-2}	5.138×10^{-3}	7.707×10^{-4}	1.245×10^{-3}
$M = 128$	2.463×10^{-2}	1.522×10^{-3}	1.921×10^{-4}	5.928×10^{-6}
$M = 256$	7.910×10^{-3}	2.461×10^{-4}	1.292×10^{-5}	4.172×10^{-6}
$M = 512$	2.212×10^{-3}	4.129×10^{-5}	5.688×10^{-7}	1.865×10^{-7}
$M = 1024$	5.830×10^{-4}	5.191×10^{-6}	1.999×10^{-8}	2.679×10^{-9}

Table 4: Errors at the starting values in the L^2 norm: $\max_{1 \leq n \leq q-1} \|u_r^s(t_n) - u_r^{bdf_q}(t_n)\|_0$.

	$q = 2$	$q = 3$	$q = 4$	$q = 5$
$M = 64$	2.911×10^{-1}	2.655×10^{-2}	7.860×10^{-3}	1.395×10^{-2}
$M = 128$	1.214×10^{-1}	7.524×10^{-3}	1.893×10^{-3}	4.806×10^{-4}
$M = 256$	3.954×10^{-2}	1.248×10^{-3}	1.217×10^{-4}	3.920×10^{-5}
$M = 512$	1.144×10^{-2}	2.160×10^{-4}	5.660×10^{-6}	2.724×10^{-6}
$M = 1024$	3.090×10^{-3}	2.770×10^{-5}	2.389×10^{-8}	4.049×10^{-8}

Table 5: Errors at the starting values in the H_0^1 norm: $\max_{1 \leq n \leq q-1} \|\nabla(u_r^s(t_n) - u_r^{bdf_q}(t_n))\|_0$.